



SEA Lab

Symbiotic Engineering and Analysis Lab



Carbon Sequestration Harvesting Energy from Waves: *Technical Narrative*

Power at Sea Prize
Department of Energy
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SEAquestration Team

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Short Description

As marine energy research continues to evolve, wave energy converters (WECs) have emerged as a promising technology for promoting grid resilience, powering remote and coastal communities, and supporting the Blue Economy. The SEAquestration team explores the integration of WECs with the growing demand for carbon sequestration, proposing a novel approach to marine carbon dioxide removal (mCDR). This approach, Carbon Sequestration Harvesting Energy from Waves, or CASHEW, seeks to directly utilize the mechanical energy produced by WEC oscillations to pump carbon dioxide into eligible geographies. The CASHEW system includes a wave energy converter, pump, and pipe. Our system design assumes that CASHEW receives pre-captured CO₂, meaning that carbon capture technologies lie outside the system boundary. In this DEVELOP phase submission, we share our refined system design, updated modeling results, and location-specific case studies that demonstrate the technical and economic feasibility of CASHEW.

1. Concept Development and Viability

CASHEW (Carbon Sequestration Harnessing Energy from Waves) is a system that uses wave energy to directly pump CO₂ into aquifers in the ocean sea bed for long-term carbon sequestration. This system, illustrated in Figure 1, will utilize wave energy converters (WECs) that will harness wave energy as mechanical energy which will then be utilized to directly drive the pumping of CO₂ via a piston. CASHEW builds upon the mission report developed by Niffenegger et al. [1]. The basic idea is that as waves excite the WEC float, the relative motion of the float drives a piston, similar to how an electricity generating WEC using hydraulics works, except instead of pumping hydraulic fluid through a motor, the piston pumps CO₂ down to the ocean floor and into the ground where it is sequestered. Current modeling has focused on the floating point absorber architecture, which oscillates in heave (this is the architecture illustrated in Figure 1), but there is also the potential to utilize the oscillating surge architecture. The piston will pump the CO₂ through a pipe that will extend from the sea surface into a seabed aquifer. The chosen aquifer must meet the requirements for either physical trapping (i.e., the presence of cap rock) or gravitational trapping (i.e., greater than 2700 meters of water depth and valid salinity and temperature profiles). While chemical trapping in basalt was a primary focus of the CONCEPT Phase, it has been deemed that it will take too long for the associated chemical reactions to occur given the system requirements on time to sequestration (see Table 1).

Location modeling using Geographic Information Systems analysis has identified two initial regions that meet system requirements: the Gulf of

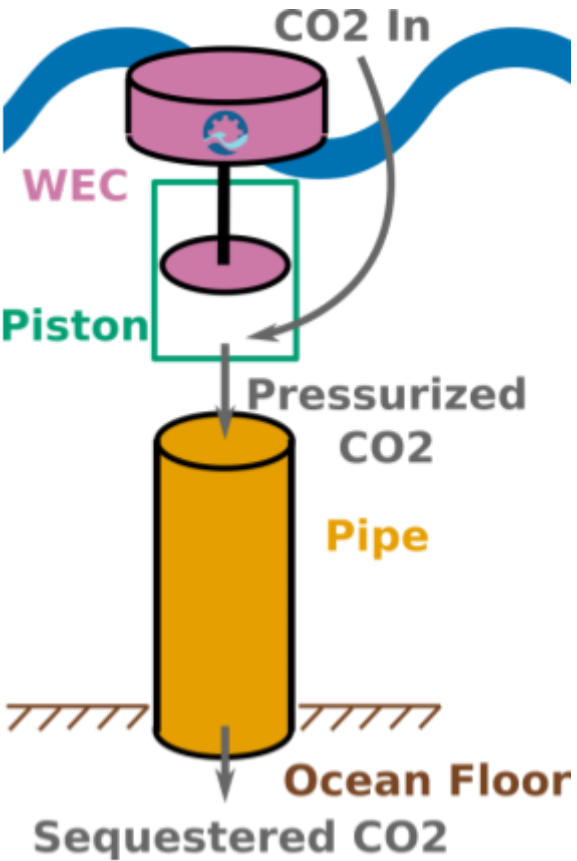


Figure 1: CASHEW system illustration.

America and the Juan de Fuca Ridge off the West Coast of the US. Under sea states relevant to these regions of interest, a single WEC can sequester 0.414 Mt/yr and 0.723 Mt/yr, respectively. Modeling of the aquifers in these regions has also indicated that they have sufficient space for sequestration on the order of the lifetime of the WEC (20 years) and more. This system lifetime helps inform a key economic metric, the levelized cost of carbon (LCOC), which informs systems-level viability as well as offers a point of comparison if this system was implemented by harnessing different renewable resources.

Because CASHEW does not operate for the primary intention of energy capture, a power matrix is not the most relevant metric to inform system performance. Instead, this system will primarily be analyzed by looking at the amount of CO₂ it can sequester per year and the levelized cost of carbon. The value for carbon sequestered per year is determined using a system dynamics model, leveraging WEC-Sim and a Simscape model of the WEC-driven piston, pipe, and aquifer. The LCOC is then calculated by combining the yearly carbon sequestration rate with the economic analysis.

1.1 Application and Challenge Area

As described in the CONCEPT Phase of the prize, CASHEW aims to address the Blue Economy application of marine carbon dioxide removal (mCDR) through the *suitability of power* challenge area. This challenge area was selected because this system looks to capitalize on the potential benefits that may be realized by coupling WECs with the process of carbon sequestration. From a geographic perspective, the regions where WECs can supply sufficient power to the system to meet both sequestration and operational demands overlap with those that have been identified as suitable for the selected sequestration method and medium (this is further discussed in the Quantitative Analysis, section 2.1, of the Early Design Document). The team's preliminary work to study this overlap has focused on two regions of interest in US coastal regions—specifically, in the Gulf of America and the Juan de Fuca Ridge—but the literature supports that there is global potential to implement a system like CASHEW [1]. Thus, scaling the system to meet the 1 Gt CO₂/yr threshold set by NREL [1] would theoretically be possible in regions that align in potential for power generation and anticipated power needs for the system. Furthermore, given that marine carbon sequestration inherently suggests a *marine-based* system for CASHEW, WECs are well suited to generate power for CASHEW because they are employed in marine applications. Thus, it would not be necessary to redesign the general architecture of the system to make it well-suited for deployment in the intended regions.

In addition to the geographical reasons, CASHEW plans to achieve suitability of power as it relates to economic concerns. WECs extract energy from waves by converting it into mechanical energy through motion. CASHEW plans to utilize this mechanical energy produced by WEC oscillations to directly drive a piston that pushes CO₂ down a pipe and into the seabed aquifer, therefore converting the energy from the waves into sequestered CO₂ without an intermediate energy conversion step. This coupling of the WEC motion with the pumping of CO₂ will enable CASHEW to significantly reduce the energy waste that would be associated with the conversion to and from electricity, thus increasing the overall efficiency of the system. Alternative renewable resource types (wind, solar, etc.) would be unable to avoid this conversion; consequently, while WECs typically have a higher LCOE (levelized cost of energy) than these alternative renewable resources, the increased efficiency makes WECs a far more attractive option to power this system. They would also eliminate the need to install the necessary infrastructure for such conversions, increasing the economic suitability and viability of power.

1.2 Concept Maturation

In the DEVELOP Phase of the prize, the system concept was matured through a more holistic literature review as well as significant strides toward more realistic modeling of the system. Before this more in-depth modeling was pursued, the team focused on clarifying the design requirements for the system, working both from a top-down (high-level systems requirements) and a bottom-up (requirements specific to various design decisions) approach. **Table 1** summarizes the finalized list of top-down requirements.

Table 1: Top-down requirements for CASHEW system

Requirement	Value
Flow Rate	1 Gt/yr of CO ₂ at scale
Time Scale	To sequestration: ~ days Stability maintained for: > 100 years
Pressure	Sufficient for desired sequestration method
Pipe Performance	Corrosion manageable during system lifetime Pipe does not fail due to hydrostatic pressure at depth
Lifetime Cost	\$6-12 per ton of CO ₂ [2]

As a result of the literature review and reformation of the system requirements, the team conducted a design review to narrow the scope of the system by defining the intended sequestration medium (where sequestration is occurring) and method (how sequestration is achieved in-situ). Moving away from fracking, CASHEW will now pursue aquifers as the sequestration medium with physical and/or gravitational trapping as the method.

With this narrowed scope for the system concept, a broad array of modeling was pursued to quantify the systems ability to meet requirements, investigating in particular flow rate, pressure, and aquifer volume. Building on the framework of the initial modeling work done in Simscape during the CONCEPT Phase, a time domain model of the system with wave dynamics modeled was created in WEC-Sim. Introducing WEC-Sim allowed for modeling the frequency dependent dynamics as well as the non-linearities in the piston and the working fluid (a two-phase fluid). WEC-Sim allows the model to capture the impact of the oscillatory nature of the wave environment and the transient dynamics. In addition to expanding the Simscape model to WEC-Sim, the modeling of aquifer dynamics was also pursued in order to determine the time-varying pressure requirements and flow rate changes associated with the filling of an aquifer. This modeling was an important verification tool to ensure that both the WECs would be able to provide sufficient power over the lifetime of the system and that the aquifers identified had sufficient space for the targeted volume of CO₂ for a given system. Then, as it relates more generally to location modeling, GIS mapping was pursued. This tool made it possible to overlay wave availability data with aquifer locations and parameters, including most importantly, permeability. GIS mapping was a crucial tool for identification and analysis of the regions that were candidates for end-use locations. Finally, throughout the DEVELOP Phase, further economics modeling helped investigate the potential benefits of the CASHEW system. As opposed to further developing a comparative LCOC metric versus other renewable energy resources as was done in the CONCEPT Phase, more time was spent developing a specific LCOC for the CASHEW system.

Throughout the DEVELOP Phase, the team leveraged Cornell University faculty as well as external resources to inform the development of CASHEW. Within Cornell, the team engaged with Dr. Mike Will and Dr. Ruth Richardson in order to better aquifer geology and dynamics. This work, alongside literature review, informed how the modeling dealt with the time-dependence of the pressure requirement in a filling aquifer and also aided in reaching the decision to move away from depleted oil wells. Beyond Cornell, the team collaborated with NREL to better understand and implement their wave data sets for the regions of interest and Dr. Jocelyn Kluger of MathWorks to improve the computational efficiency of the system dynamics model.

2. Blue Economy Application Evaluation

As marine energy research continues to evolve, WECs have emerged as a promising technology for promoting grid resilience, powering remote and coastal communities, and supporting the Blue Economy [3]. Wave power, while still intermittent, has significantly greater energy density than its renewable counterparts like solar and wind [4]. CASHEW seeks to utilize wave power's high energy density to sequester carbon dioxide. CASHEW seeks to utilize direct drive—the process by which mechanical energy as a result of WEC oscillations is used to directly pump captured CO₂ into identified saline aquifers. Using direct drive has the potential to reduce energy losses associated with converting mechanical to electrical energy [5], thereby making it both more efficient and scalable.

As a method of WEC-powered sequestration, mCDR serves as our selected primary blue economy application. To achieve this application, a variety of requirements will need to be determined. We also discuss how CASHEW will meet mCDR needs, as well as how competitive it is compared to existing mCDR companies.

2.1 CASHEW Meeting Blue Economy Application Requirements

To meet our sequestration goal of 1 Gt CO₂/year at scale, the CASHEW system needs to meet the top-down requirements described in *Concept Development & Viability*. If met, CASHEW will, in theory, meet minimum requirements to function and potentially be scalable. To meet these requirements, we can investigate the following categories:

1. Sequestration Method: As discussed in *Concept Development & Viability*, CASHEW aims to use either physical or gravitational trapping, depending on conditions of identified locations.
2. Potential Location of CASHEW System: As developed in the Early Design Document, potential locations for CASHEW deployment are the Juan de Fuca Plate (suitable for both physical and gravitational trapping), or the Gulf of America off the coast of San Jose, Mustang, and Padre Islands (suitable for physical trapping).
3. CASHEW Power Requirement: Since CASHEW aims to sequester 1 Gt CO₂/year through WECs, the primary objective is to determine how much power this would require. As investigated in the Early Design Document, available wave power in Juan de Fuca is an average of 61.8 kW/m, while in the Gulf, it is an average of 1.6507 kW/m.
4. Cost: To be cost-competitive with existing carbon capture technologies, CASHEW aims to cost between \$6-12/tCO₂. Currently, our economic model suggests that the system's LCOC is \$6.30/tCO₂ or \$10.21/tCO₂ depending on the location.

2.2 CASHEW Compared to mCDR Existing Technologies

mCDR encompasses a wide variety of approaches such as seawater electrolysis, mineral weathering, ocean alkalinity enhancement, microalgae cultivation, etc. Each of these approaches aims to remove carbon dioxide from the atmosphere and store it via an ocean-based solution. CASHEW, unlike many of WEC-based technologies, does not aim to produce electricity, but rather to use the mechanical work from wave oscillations to pump CO₂, thereby sequestering it. Within the marine carbon sequestration (mCDR) space,

technologies like Ebb Carbon’s Electrochemical Ocean Alkalinity Enhancement and Equatic’s Electrolysis-Based CO₂ Removal & Hydrogen Production [6] are existing technologies.

Table 2: Overview of various carbon capture technologies.

Company	mCDR Technology Used	Method
Equatic [7]	Electrochemical Direct Ocean Capture (DOC)	Uses electrolyzer (powered by renewables) to split seawater into acidic and alkaline streams as well as streams of hydrogen and oxygen. The alkaline stream captures atmospheric CO ₂ , transforming it into stable bicarbonates and carbonates, which are released back into the ocean. Hydrogen is captured to be used as green hydrogen.
Ebb Carbon [6]	Electrochemical Ocean Alkalinity Enhancement	Separates seawater into acidic and alkaline streams, returning the alkaline stream to the ocean. This increases ocean alkalinity and thereby promotes CO ₂ removal.
Captura [7]	Direct Ocean Capture (DOC)	Uses electrodialysis and gas extraction technologies to captures CO ₂ directly from seawater
Vesta [8]	Mineral Weathering	Spreads ground olivine on coastlines to naturally enhance ocean alkalinity, capturing CO ₂ .
Planetary [9]	Ocean Alkalinity Enhancement	Introduces alkaline materials into the ocean to increase alkalinity, promoting CO ₂ removal.

As demonstrated in **Table 2**, most existing mCDR companies rely upon ocean alkalinity enhancement or electrochemical solutions. While these methods have been studied and are currently deployed, many of these methods require altering the ocean’s alkalinity, which could have potential adverse effects. Using CASHEW, ocean chemistry can remain relatively unchanged. Moreover, CASHEW could potentially be developed to include energy production in the event that there is significant energy surplus (i.e., beyond what is necessary for sequestration). This means CASHEW has the potential to function as both a method of mCDR, as well as a source of additional energy at sea. This, coupled with CASHEW’s direct-drive approach, make it uniquely suited to efficiently address sequestration power needs. That being said, it is important to note that CASHEW is a significantly lower technological readiness level compared to technologies discussed in **Table 2.**, as CASHEW is novel. In a similar vein, CASHEW is projected to cost more than technologies like mineral weathering, as this method does not have significant components, and therefore limited expenses (i.e., primary expense in mineral weathering is olivine).

2.3 WEC-Driven vs WEC-Powered

While CASHEW addresses mCDR as its Blue Economy application, given its unique approach to mCDR (wave power), it is important to compare the system to other WEC power takeoff systems. Our approach is advantageous as it directly uses mechanical energy in its power takeoff system (PTO). Other takeoff systems, however, include hydraulic PTO, where WEC oscillations are used to power a hydraulic motor, which drives a generator, or electrical PTO, directly converts linear motion of the wave energy device into electrical power via magnets and coils. In either case, there is a power efficiency of 74%-80% [10], or 80.6%, assuming a linear switched reluctance generator [11]. In both cases, there are power losses that could be mitigated by using direct drive, as CASHEW aims to.

In addition to discussing other PTOs, we can contextualize how WEC-driven pumping is advantageous compared to other methods of CO₂ pumping. Currently, most pumping-related CO₂ sequestration is operated by the oil and gas industry. CO₂ is often compressed into a supercritical state, and is then pumped into reservoirs. These pumps are typically powered by electric motors using power taken from the electricity grid, natural gas turbines, or diesel engines. In any case, power is either generated using fossil fuels or requires energy from the grid. CASHEW is novel in that it produces the energy it will need to achieve sequestration—not take it. Given this self-sufficiency and potential ability to generate electricity in addition to pumping CO₂ at scale, CASHEW has potential environmental and economic benefits that differentiate it from its peers.

3. End User(s) Identification

CASHEW has promising advantages for several key players in the decarbonization, energy, and sustainability spheres. Significant end users include corporations, political leaders, environmental agencies, and scientific researchers. As carbon markets continue to expand, governments and corporations are interested in trading carbon credits alongside their transition to low-carbon technologies to maintain an overarching emissions limit. Many corporations are also investing in carbon offsets to comply with

environmental regulations and counteract greenhouse emissions that may result from their activities. Through effectively sequestering carbon dioxide in the ocean, CASHEW can thus be a new tool to achieve these goals for stakeholders in the carbon tax market.

Furthermore, political leaders can benefit from incorporating CASHEW to help meet their countries' national decarbonization goals. By approving the implementation of the system, government officials can deploy another method of reducing carbon emissions. With a competitive levelized cost of carbon, the system is cost-effective and a viable tool for political leaders to consider. The marine-based nature of CASHEW is particularly advantageous for these end users. Applying such a platform for carbon capture purposes additionally minimizes land use and helps conserve space onshore compared to traditional direct air capture systems, which is a crucial advantage given increased urbanization, population growth, and demands for housing and infrastructure in recent years.

Similarly, environmental agencies such as the US Department of Energy can meet their objectives related to deploying sustainable energy systems and mitigating emissions through CASHEW. Scientists and engineers working in the intersection of marine systems, sustainable energy, and carbon capture will also be able to incorporate real-world data from the CASHEW system in their research, spurring the development and optimization of innovative energy solutions.

CASHEW additionally has the potential to facilitate enhanced oil recovery (EOR) as the carbon dioxide sequestered by the system can be integrated with EOR processes to increase extraction efficiency. In doing so, this can facilitate oil production while minimizing atmospheric CO₂ emissions, making CASHEW an attractive option for oil and gas companies. From a financial standpoint, revenue earned from applying EOR can also offset expenses associated with CASHEW, improving the economic viability. The system can thus be a useful component of the energy transition.

4. Future Plans

4.1 Plans for Advancement

The first step of the plan for advancement is to further refine the model. The economic model is not quite complete and does not capture all the design tradeoffs between locations. Adding more considerations on PTO costs and quantifying possible savings through co-locating with decommissioned offshore oil and gas platforms, would allow for a more thorough understanding of optimal locations and design. Additionally, we started work on a multi-fidelity lifetime performance assessment of the CASHEW system that is not quite complete yet. In Section 2.2 the Early Design Document, we describe the system dynamics model we built to simulate the CASHEW system's performance on the order of minutes. This model can be used to evaluate the performance of the system at a moment in time allowing us to estimate the rate of sequestration, but the time scale is too small to understand the impact of changing aquifer parameters. As CO₂ is sequestered into an aquifer, the aquifer fills up, increasing the pressure the system needs to overcome [12]. This happens over a much larger time scale though. We are interested in first building a model of how the pressure in the aquifer changes over time, and then integrating it with the system dynamics model we have already. This would enable us to simulate the full lifetime performance of the CASHEW system by running the system dynamics model for a few minutes, using the resulting average flow rate as the input to the aquifer filling model, running that for a few months, updating the system dynamics aquifer parameters, and iterating. This is shown graphically in **Figure 2**.

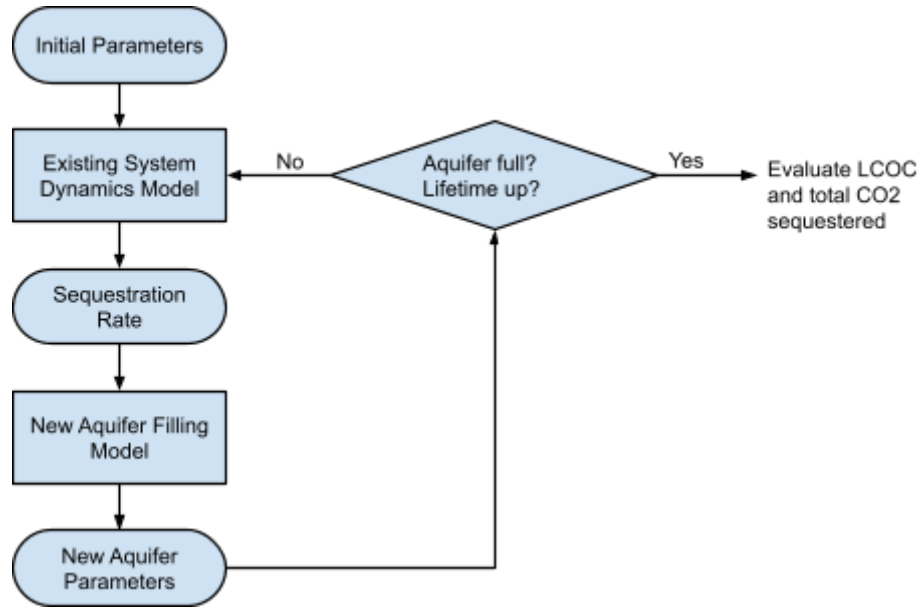


Figure 2. Multi-fidelity CASHEW model flow chart.

Beyond improving the CASHEW modeling, we also plan to put together a physical proof of concept prototype. The proof of concept prototype will consist of a WEC, a WEC-driven hydraulic cylinder, and an artificial aquifer. The artificial aquifer will be created using a permeable membrane with its permeability and thickness selected to properly scale with the WEC size and choice of working fluid to model the CO₂. We will aim to design, build, and test this prototype all within the next 12 months.

The budget for these next steps is estimated at \$75,000. The full cost breakdown can be found in the Early Design Document. Given this, winning this prize would enable our ability to complete this project, without needing to reach out to outside resources for additional funding. However, more funding would give us more flexibility and opportunities to improve the prototype design. Our lab has had a significant history of success winning funding from within our institution for offering undergraduate researchers paid positions and acquiring additional funds for prototyping projects. In the past, we have had great success applying for and receiving funding from the Atkinson sustainability center at Cornell, and this project would be a great fit.

Beyond the next year, another major way to grow this project is through optimization. Notably, the WEC design was not at all optimized, and just taken from literature [13]. Improving the WEC design and piston design through optimization could have a significant positive impact on system performance. Our lab has a lot of experience in optimization of WEC systems, including the optimization of WEC design and PTO together through Multidisciplinary Design Optimization (MDO) [14,15]. In one notable study, using MDO resulted in 60% improvement in system performance [16].

4.2 Team Gaps

There are two major knowledge gaps in the current team given the future direction of this project. First, is the lack of a specialist for understanding aquifer dynamics. Cornell has a department of Earth and Atmospheric Science that we will recruit a student from to aid in our understanding of aquifer dynamics. The student will be well versed in the geologic language that we are still getting used to and also have easier access to resources for helping to understand these dynamics. This will aid in the further development of our model as well as the design of a lab “aquifer” for a proof of concept model.

The second knowledge gap is in experimental fluids experience. Our lab has experience prototyping and testing WECs and has experience with Froude scaling. However, our lab has less experience with data analysis of experimental fluids data. To address this gap, our team will recruit a student who has more experimental fluids experience as well as have some of the team take the experimental fluids course (CEE 6270) offered at Cornell. Addressing this gap will enable the team to move forward with the prototyping direction to develop a proof of concept model.

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